

I am a Human-Centered Computing researcher focused on advancing Mixed Reality (MR) by measuring, interpreting, and adapting key system parameters and services to improve user immersion, safety, and overall experience. My research focuses on designing the next generation of adaptive Human-Centered MR by aligning technologies with user experiences and integrating human factors into every aspect of MR design and development, using tools and techniques from Systems, Human-Computer Interaction (HCI), and Machine Learning (ML).

My contributions have been recognized through high-impact publications, open-source datasets [1], and follow-up collaborations with industry and academia. I have worked across institutions, including UMass, MIT, UMN, UTAH, UC Irvine, WPI, Microsoft, and StegAI. These collaborations span technical specialties beyond my primary research areas, such as computer graphics, computer vision, control systems, distributed systems, and networks. This interdisciplinary exposure positions me to lead multi-institutional projects that align with the priorities of major funding agencies. My CRA-E Fellowship experience further enhanced my ability to articulate the broader impact of my work, engage with funding agencies and secure resources to advance my goals.

My research has received broad recognition, including being named an MIT EECS Rising Star, CPS Rising Star, ML and System Rising Star, Young Research at Heidelberg Laureate Forum, and CRA-E Fellow, as well as a Grace Hopper Scholar and Google CSRMP Scholar. I was a finalist in UMASS 3MT and AIxVR and received the Best Presentation Award at the ACM SenSys PhD Forum. Additionally, my commitment to fostering inclusive research environments was recognized with the UMass College of Engineering DEI Award. These honors reflect my research contributions and my dedication to mentorship, collaboration, and broadening participation in computing.

Research Summary

MR combines physical and virtual worlds by sensing the user's environment and interactions, using user position tracking to align virtual elements with real-world objects accurately. The system processes this user position tracking data so virtual objects respond naturally to the user's actions in real-time, creating an immersive experience. This immersion creates the experiential aspect of MR: presence, the feeling of "being there" in the virtual environment (VE). MR applications place unique demands on both user experience and the underlying technology (such as tracking and real-time responsiveness), requiring the system to adapt. For instance, gaming prioritizes fast response times and high visual quality, while surgical applications require accurate tracking during critical procedures. In multi-user collaborative settings, the system must carefully manage what to share and when, balancing resources, security, privacy, and fairness to prevent a single user from disrupting the experience for others. Given the unique demands of MR applications, which must handle the unpredictability of human interactions and adapt to changing conditions, whether related to the environment, user behavior, collaboration, or resource constraints, my work focused on three main thrusts:

① Rethinking How We Sense Presence in Real-Time (VR x 3, TVCG, SafeAR@ISMAR).

Experiential Thrust laid the scientific basis for understanding immersive experiences in MR. I identified the link between human cognitive state and their performance [2]. I also developed objective, quantifiable measures of presence [3], an important phenomenon that dictates human behavior and task performance in VE. These foundations guide how we build and evaluate future MR systems, shaping the design of interactions to align with cognitive processes and user perceptions of the VE [4]. Traditional methods relying on subjective questionnaires miss real-time, nuanced responses. I proposed new adaptable system metrics, such as reaction time as

the measure of presence. My approach enabled systems to adapt to users' cognitive states and sensory feedback dynamically to a range of factors [5], including user cognitive states [2], system fidelity [3], interaction design [4], and real-world disruptions [6] while maintaining immersion.

② Scene-Driven Adaptive Human-Centered Tracking (*IROS, AIXVR, DATA@SenSys*).

Systemic Thrust contributed to core technological infrastructure, improving tracking accuracy, adaptability, and security in MR. Accurate, real-time tracking is essential for MR, yet current systems struggle to adapt to rapidly changing environments and unpredictable human interactions, leading to cognitive overload and reduced immersion. To address this, my research introduced the Neurosymbolic Feature Extraction (nFEX) framework [7] that integrates a taxonomy of human interactions and environmental factors [8] into the tracking process. To support nFEX, I developed the HoloSet dataset [1], providing essential data for underexplored MR scenarios. Additionally, I am developing ContinuumTrack [9] to anticipate and resolve tracking failures, ensuring an uninterrupted user experience. By addressing vulnerabilities in MR systems [10], my work further boosts adaptability and security to improve the overall user experience.

③ Learning to (Safely) Collaborate (*NeurIPS, FAccT, ISEMV, AIChallengesT@SenSys*).

User Collaboration Thrust focused on fairness, privacy, security, and resource efficiency in shared MR experiences to promote equitable interactions between users. My contributions included developing techniques to sense [11] and interpret individual user behaviors [12], allowing MR systems to infer group dynamics to optimize interactions [13]. I introduced methods to mitigate biases [14–16] and enhance privacy and security [17–19] through Federated Learning (FL), to support secure and fair data handling across multiple users. Additionally, by optimizing resource efficiency to balance quality-of-service (QoS) with quality-of-experience (QoE) [20], my work supported the sustainability of MR systems [21]. This aligns technological innovation with broader societal goals, such as decarbonization and resource conservation.

Future Research Directions

My future research will explore how we can learn from users as they interact with systems to create personalized experiences in MR and adjacent human-centered systems by integrating user behavioral state estimation and physical tracking with collaborative model training. My research will balance personalization with safety, well-being, fairness, and societal goals such as mitigating climate change and improving health. Grounded in an extensive theoretical background and human subjects research, I will introduce new tools, services, control parameters, and novel uses of existing data, creating adaptive and personalized experiences across diverse environments and hardware platforms.

Broader Impact.

While my research advances MR systems, it has far-reaching societal implications. The behavioral and safety tools I develop in my experiential and systemic thrusts have applications in accessibility and healthcare, particularly for obesity, diabetes, and aging, aligning with National Institutes of Health (NIH) goals. My real-time behavioral monitoring tools can be applied to monitoring behaviors in neurodivergent individuals or those with autism for personalized support, tracking cognitive and physical activity changes in aging to enable early interventions, and monitoring activity and diet to promote healthier choices in diabetes management¹. Finally, with my interdisciplinary background, I can apply to and find both the NSF and NIH CAREER programs highly relevant to my research trajectory.

¹NSF IIS:HCC, NSF CCF:FET, NIH BRAIN Initiative, NIH AI and Technology Collaboratory, NIH VR Technologies for Obesity and Diabetes

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